

Minicurso: **Ecosistemas sob uma nova perspectiva (SIG e Sensoriamento Remoto)**

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Guia de Preparação e Organização

Olá estudante! Para que a nossa aula prática flua perfeitamente e sem erros inesperados, é fundamental que você prepare o seu ambiente de trabalho seguindo este guia rápido.

1. Organização de Pastas (Obrigatório)

Softwares de Sistema de Informação Geográfica (SIG), como o QGIS, costumam apresentar erros se os arquivos estiverem em pastas com caminhos muito longos ou com nomes mal formatados.

O que você deve fazer:

1. Abra o **Explorador de Arquivos** do seu computador.
2. Vá até o disco local principal, geralmente o **Disco Local (C:)**.
3. Crie uma pasta nova diretamente no disco C.
4. Nomeie esta pasta como: **Minicurso_SIG** (O caminho final ficará: C:\Minicurso_SIG).

 **Importante:** Salve todos os arquivos, projetos e resultados gerados durante o minicurso exclusivamente dentro desta pasta. Não use a Área de Trabalho (*Desktop*) ou a pasta *Downloads*.

2. Boas Práticas na Nomeação de Arquivos

Ao trabalhar com dados espaciais e programação, o nome que você dá aos seus arquivos pode ser a diferença entre um trabalho com processamento fluido e em menos tempo, e uma sequência de erros ou maior tempo de processamento.

Lembre-se sempre dessa regra: **Nunca use espaços ou caracteres especiais.**

-  **Incorreto:** bacias da amazônia versão final.shp (Tem espaços, acentos e caracteres

especiais).

-  **Incorreto:** áreas_prioritárias_ç.geojson (Tem acentos e cedilha).
-  **Correto:** bacias_amazonia_vfinal.shp (Usa *underline* no lugar de espaços, sem acentos).
-  **Correto:** areas_prioritarias.geojson (Limpo e direto).

3. Download da Base de Dados

Todos os arquivos que usaremos na prática serão disponibilizados à você em um repositório dedicado ao minicurso em meu perfil do GitHub. Basta seguir os seguintes passos:

1. Acesse o repositório oficial da aula no GitHub:  [Acessar Base de Dados no GitHub](#)
2. Na página do GitHub, leia as orientações gerais para identificar quais bases de dados deverão ser baixadas e qual o formato recomendado.
3. Após baixar, **extraia (descompacte)** todos os arquivos do ZIP diretamente para a pasta C:\Minicurso_SIG que você criou no Passo 1.

4. Instalação do QGIS

O QGIS é o software livre e de código aberto que usaremos na primeira parte da aula. Caso você queira utilizar seu computador pessoal e ainda não tem acesso ao software, siga as instruções abaixo para baixá-lo:

1. Acesse o site oficial do QGIS:
 <https://qgis.org/>
2. Clique no botão de "**Baixar Agora**" (**Download Now**).
3. Escolha a versão recomendada para o seu sistema operacional (Windows, macOS, etc.).
 - *Dica:* Recomendo baixar a versão **LTR (Long Term Release / Lançamento de Longa Duração)**, pois ela é a mais testada e estável.
4. Após o download, execute o instalador e siga o passo a passo padrão (Next > I Agree > Install). Não é necessário alterar as configurações padrão de instalação.

5. Acesso ao Google Colab (Programação em Nuvem)

Ao final do minicurso, usaremos o Python para analisar nossa base de dados. A grande vantagem é que não precisaremos instalar nada no computador, pois faremos isso na nuvem do Google.

Requisitos:

- Uma conexão com a internet.
- Uma conta ativa do Google (pode ser o seu @gmail.com pessoal ou conta institucional vinculada ao Google).

Passo a passo para acesso:

1. Acesse a plataforma pelo link:
 <https://colab.research.google.com/>
2. Caso não esteja logado, clique em "**Fazer login**" no canto superior direito e entre com sua conta do Google.
3. Se aparecer uma janela pop-up, você pode fechá-la por enquanto. O ambiente está pronto para uso!
4. Durante a aula, nós clicaremos em *File > New notebook* (Arquivo > Novo notebook) para começar a programar.

Após seguir as recomendações, tudo estará pronto para a nossa prática! 🚀

Recomendo que explorem os softwares que serão utilizados e, para isso, as ferramentas de Inteligência Artificial podem ser uma ótima ferramenta para auxiliar nesse primeiro contato. Faça testes e anote suas dúvidas para debatermos em aula.

Falando em inteligência artificial, neste mesmo documento eu anexei 4 artigos que debatem sobre o uso dessas ferramentas no ambiente acadêmico discutindo sobre suas vantagens, limitações e recomendações. Sugiro uma leitura rápida para orientar o seu estudo.

Por enquanto é isso, nos vemos na aula prática!

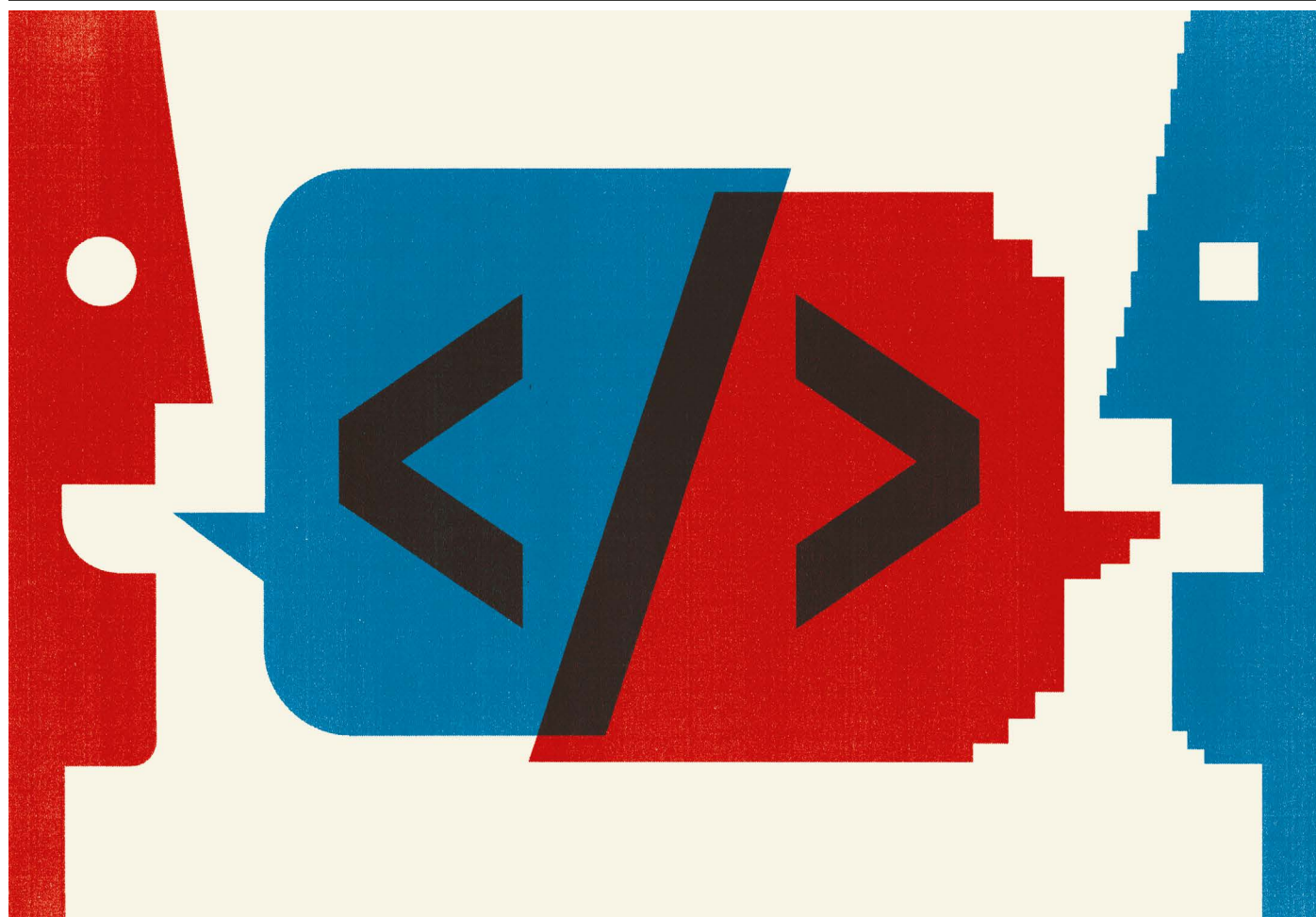


ILLUSTRATION BY THE PROJECT TWINS

SIX TIPS FOR BETTER CODING WITH CHATGPT

Although powerful, the tools are not as intelligent as they seem. Use them with caution, computer scientists warn. **By Jeffrey M. Perkel**

Unless you've been living under a rock, you know about ChatGPT. The chatbot, driven by artificial intelligence (AI) and created by OpenAI in San Francisco, California, provides eerily human-like responses to user questions (called prompts) on almost any subject. ChatGPT is trained on a vast corpus of text, and its ability to engage in text-based conversation means that users can refine its responses. Even if its initial answers are wonky, it often eventually produces accurate results, including software code.

Researchers can use ChatGPT to debug and annotate code, translate software from one programming language to another and perform rote, boilerplate operations, such as plotting data. A March preprint reported that the program could solve 76% of 184 tasks in an introductory bioinformatics course, such as

working with spreadsheets, after a single try and 97% within seven attempts¹.

That's good news for researchers who feel uncomfortable coding, or who lack the budget to employ a full-time programmer – for them, chatbots can be a democratizing tool.

Yet for all their apparent sentience, chatbots are not intelligent. They have been called stochastic parrots, randomly echoing back what they've seen before. Amy Ko, a computer scientist at the University of Washington in Seattle, invokes a long-running US quiz show to describe the tool's limitations, writing on the Mastodon social-media site: "ChatGPT is like a desperate former Jeopardy contestant who stopped following pop culture in 2021 but really wants to get back into the game, and is also a robot with no consciousness, agency, morality, embodied cognition, or emotional

inner life." (The data used to train ChatGPT only extend into 2021.)

In short, ChatGPT and related tools based on large language models (LLMs), which include Microsoft Bing and GitHub Copilot, are incredibly powerful programming aids, but must be used with caution. Here are six ways to do so.

Choose your applications

Chatbots work best for small, discrete programming tasks, such as loading data, performing basic data manipulations and creating visualizations and websites. But that's not the same as software engineering, says Neil Ernst, a computer scientist at the University of Victoria in Canada.

"Software engineering is a lot more than just solving a programming puzzle," Ernst explains. "There's thinking about test frameworks,

writing maintainable code and understanding the trade-offs involved in building a system” – such as that between speed and readability. “I don’t think that the current tools are solving any of those issues.”

That leaves many tasks they can do, such as remembering the syntax for creating visualizations with Matplotlib, a graphing library for the programming language Python. In that sense, chatbots are like a conversational interface to Stack Overflow, an online question and answer forum for programmers. “That’s not stuff that anyone particularly enjoys writing,” says Ernst, “and it saves time for us to ask the tough analytical questions that we might have about the data.”

Chatbots are also good at explaining why code doesn’t work. Emery Berger, a computer scientist at the University of Massachusetts Amherst, has exploited those abilities to build several helpful tools. One, called *cwhy*, uses ChatGPT to explain compiler errors in code written in the programming languages C, C++ and Rust. Another, ChatDBG, provides a conversational interface for debugging, and a third, Scalene, uses the AI to suggest code optimizations to improve performance.

Chatbots can even translate code from one programming language to another. Mathieu Coppey, a biophysicist at the Curie Institute in Paris, is using ChatGPT to help him move from MATLAB, his preferred language, to Python. Using Google and online forums, he typically needs days to get his Python code working. “Now, I can do that in an hour or so,” he says.

Trust, but verify

Chatbots might not always know what they’re talking about, but they certainly sound like they do. In some cases, the AI doesn’t understand the question; at other times, it provides an incorrect answer. When the code fails to run, such mistakes are obvious. Sometimes, however, the code runs but yields the wrong result.

According to a study² co-authored by linguist Emily Morgan at the University of California, Davis, chatbots – like the human-written code on which they were trained – often create what she calls “simple, stupid bugs”. These single-line errors, such as using `>` instead of `>=` in a conditional statement, are easy to fix, but hard to find. “If you don’t know enough to tell the difference between something correct and something that’s effectively nonsense, then you could get yourself in trouble,” she says.

Iza Romanowska, a complexity scientist who studies ancient civilizations at the Aarhus Institute of Advanced Studies in Denmark, has used ChatGPT to produce code in a language called NetLogo. Because there’s less online code written in NetLogo than in the languages Python and R, ChatGPT is less fluent in it. Sometimes, the AI peppers its suggested code with functions that don’t actually exist, she says – a

behaviour sometimes called hallucination.

The bottom line is not to blindly accept what ChatGPT gives you – read it carefully and test it. Make sure it performs as expected on ‘edge cases’ – for instance, does an algorithm to sort n numbers include the n th number? Patrick Lam, a computer scientist at the University of Waterloo in Canada, says: “I wouldn’t trust this further than I can throw it.”

Think safety

Chatbots output code that reflects their training data. That’s not always a good thing, says Ko. “The aggregate quality of code on the web that’s shared, that these [chatbots] are trained on, is actually quite low.”

Just as random code online is unlikely to be particularly efficient or robust, so too is chatbot-generated code. It might not work well on large data sets, for instance, and can contain security vulnerabilities.

“I wouldn’t trust this further than I can throw it.”

Brendan Dolan-Gavitt, a computer scientist at New York University, says that when Github’s Copilot programming tool launched in 2021, he and his team tested it in 89 security-relevant scenarios. One was the ability to check for malformed queries using the language SQL that could corrupt a database – known as an SQL-injection attack³. “About 40% of the time, Copilot was producing code that was vulnerable.” That’s a moving target – when Dolan-Gavitt put those scenarios to a newer version of the LLM underlying ChatGPT, called GPT-4, the error rate fell to 5%.

Still, it pays to check your code. But also consider the application – not everything is mission-critical. The web interface to a database or visualization tool, for instance, might require extra vigilance. But if you know what the answer to your programming problem should look like, “Just go for it”, says computer scientist Sayash Kapoor at Princeton University in New Jersey, “because it’s easy to check if you’re wrong.”

Iterate

Chatbot-based coding, says Ko, “is not a single-shot sort of experience”. It’s a conversation. “You write something, you get something back, you read it sceptically, you ask for more detail, you ask for it to fix something.”

Gangqing (Michael) Hu, who runs the bioinformatics core facility at West Virginia University in Morgantown, capitalized on that iterative workflow to develop a method that beginners in bioinformatics can use to optimize chatbot prompts, called OPTIMAL⁴. Users provide detailed prompts, test the replies and feed back to the chatbot to tweak

its responses. That can include questions about errors as well as tweaks to the prompt itself. “Communication is the key,” Hu explains.

If you get stuck, try adjusting the settings, suggests Xijin Ge, a bioinformatician at South Dakota State University in Brookings. The ChatGPT ‘temperature’ setting, for instance, controls creativity – the higher the temperature, the more creative the output. “Sometimes it works,” Ge says.

But not always – in some cases, “you’ll have to intervene and take over”, says Ko.

Anthropomorphize

Chatbots aren’t people, but it can be helpful to treat them that way. “Treat this AI as a summer intern,” Ge advises – a college student who is hard-working and eager to please, but also inexperienced and error-prone.

Avoid ambiguity, and break your problem up into smaller pieces, suggests Paul Denny, a computer scientist at the University of Auckland, New Zealand.

Another tip: direct the chatbot to assume a role, such as a biologist who is fluent in Python. Specify the tools or programming libraries you would like to use. Such directives can help the chatbot to get “into the right probabilistic space”, says Ko – that is, home in on the text that is most likely to follow the prompt.

For instance, one prompt in Hu’s study⁴ asked ChatGPT: “Act as an experienced bioinformatician proficient in ChIP-Seq data analysis, you will assist me by writing code with number of lines as minimal as possible. Reset the thread if asked to. Reply “YES” if understand.”

And, if possible, provide starting code, comments and expected results. “Examples can really help ChatGPT to target it in the right direction,” says Dong Xu, a computer scientist at the University of Missouri, Columbia.

Embrace change

Finally, LLMs are constantly evolving, and becoming more powerful. That’s good news for researchers, although it will keep them on their toes. Prompt lengths are increasing, allowing for more nuanced responses. And new tools are constantly emerging. One plug-in called Code Interpreter turns ChatGPT into a digital data analyst, allowing users to upload data sets, ask questions of their data and download results. As one blogger on AI put it, “It’s like having a conversation with your data. How cool is that?”

Jeffrey M. Perkel is technology editor at *Nature*.

1. Piccolo, S. R. *et al.* Preprint at <https://arxiv.org/abs/2303.13528> (2023).
2. Jesse, K. *et al.* Preprint at <https://arxiv.org/abs/2303.11455> (2023).
3. Pearce, H. *et al.* Preprint at <https://arxiv.org/abs/2108.09293> (2021).
4. Shue, E. *et al.* *Quant. Biol.* <https://doi.org/10.15302/J-QB-023-0327> (2023).

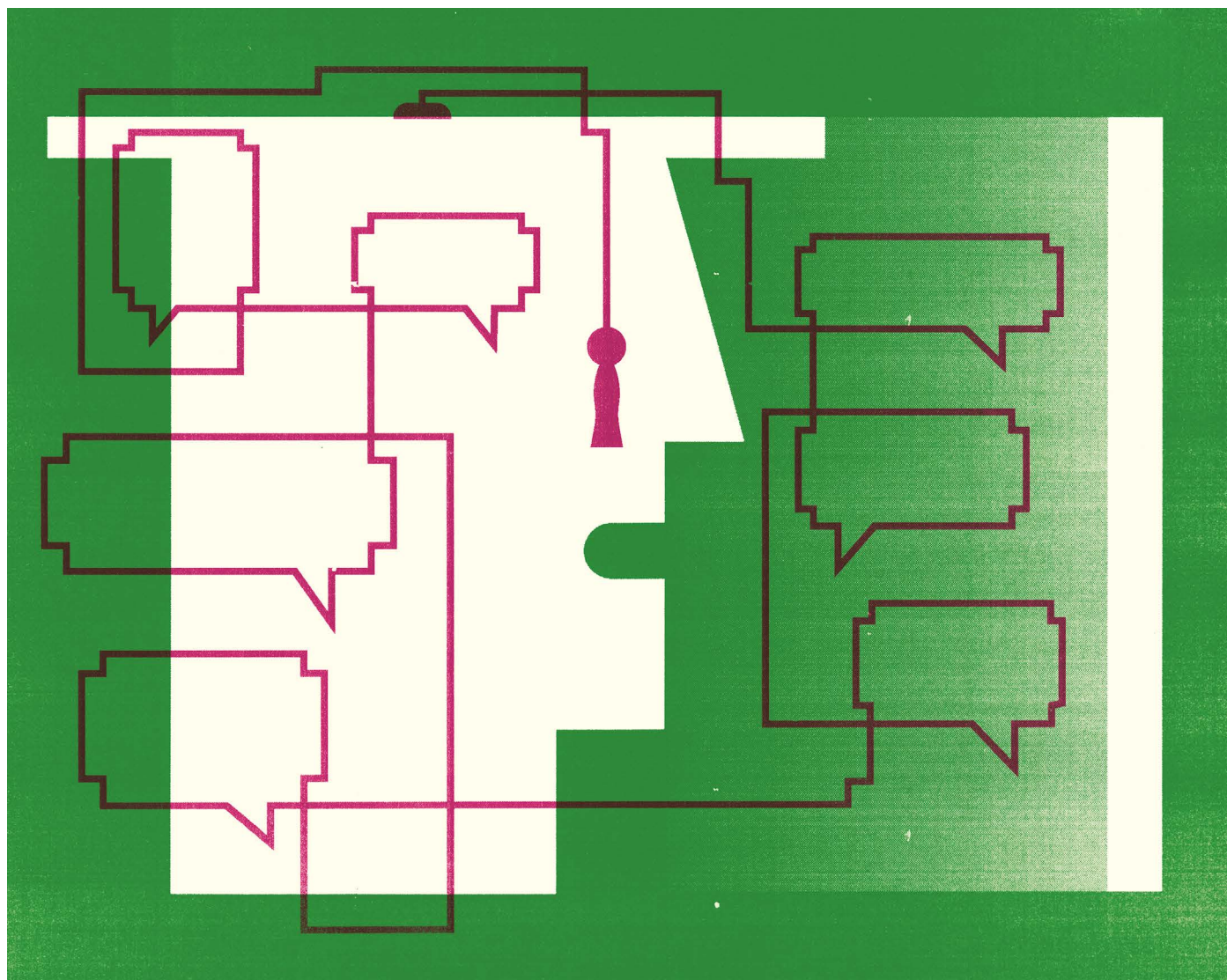


ILLUSTRATION: THE PROJECT TWINS

STUDENTS FIND NEW USES FOR CHATBOTS

The use of generative AI tools has gone beyond simple summarization and grammar – students are now learning from AI professors. **By Amanda Heidt**

By the time her comprehensive exams rolled around in August 2024, Adriana Ivich had done nearly everything she could to prepare herself. To officially become a PhD student in biomedical informatics at the University of Colorado Anschutz Medical Campus in Aurora, Ivich needed to present her proposed research project and then meet with her committee for a closed-door grilling.

“It’s definitely one of the hardest times to be a PhD student,” Ivich says. “I spent months – years really – preparing, but you still don’t know what they’re going to ask you.”

She could make an informed guess, however. Months before, she’d poured the biographies and publications of her five committee members into ChatGPT, a generative AI chatbot developed by OpenAI in San Francisco, California, and used it to create digital simulacra of each person. She then fed the mimics her research proposal and asked the program to respond as her committee might, asking detailed questions and identifying gaps in her knowledge. The actual meeting “went pretty much like ChatGPT said it would,” says Ivich, who passed with flying colours.

Her experience highlights just one of the

many imaginative ways in which students are using generative AI. Unlike the ‘early days’ of two years ago, when using AI meant summarizing a paper or outlining an essay, students are now leaning into the tools’ ability to emulate human connection, turning chatbots into podcast hosts, language tutors, professors and even personal trainers.

“I want professors to work with students assuming they’ll be using these models,” says Lily Lee, who is studying manufacturing and design engineering and music at Northwestern University in Evanston, Illinois, and is co-creator of the newsletter AI × Education.

Work / Technology & tools

Although some educators worry about the erosion of critical-thinking skills, she says, “another way to look at it is that AI can help students learn more creatively, which helps them build their critical thinking as well.”

A back-and-forth conversation

When ChatGPT first appeared in November 2022, many educators and academic institutions responded with blanket bans, unsure whether these tools were more likely to spark innovation or cheating. The jury is still out on that, but administrators have started to relax some policies and give teachers greater freedom to direct when they can be used. How these tools are used, however, is often driven by students.

“I’m pleasantly surprised at some of their creative uses,” says Victor Lee, a learning-sciences researcher at Stanford University in California who focuses on the intersections between AI and education.

Like Ivich, many students seem drawn to applications that showcase the ‘human’ side of programs such as ChatGPT and NotebookLM, an AI-powered interface from Google in Mountain View, California, for interpreting documents. Although early iterations of these tools required users to key in queries, newer versions allow people to speak into a microphone, enabling more-natural, organic conversations.

Sami Melhem, who is studying computer science at Texas A&M University in College Station, created his own chatbot, Class Primer, to tutor him. Built atop ChatGPT-4, the program begins by asking for details about a class, including a course description, learning outcomes, required textbooks and a syllabus. Then, it breaks down the course into ‘priming sessions’ spread out over the semester.

Melhem now spends his Sunday afternoons familiarizing himself with a few weeks’ worth of material before the concepts are introduced in class. The bot lets Melhem engage with his lectures in different ways, such as by creating playful analogies or visual maps that link concepts and show how they connect over the semester. As a visual learner, Melhem says that the diagrams especially “help with understanding the subject as a whole”.

Other students, such as Jan Bartkowiak, who is studying economics at Minerva University in San Francisco, California, have made chatbots that emulate famous historical figures to interrogate their thinking. In 2024, Bartkowiak wrote an essay on the Vietnam War and used this exercise to probe how historians or contemporary political figures, such as Henry Kissinger, who served as the United States national security adviser and secretary of state during the conflict, might have framed certain events.

“You might want to quickly identify which book or philosopher supports your point, so

instead of starting your research by digging deeply into those books, you can start with ChatGPT,” he says. “You can have back-and-forth conversations and ask clarifying questions, and that’s been very useful.”

This ability to speak to chatbots has proven particularly helpful for students taking courses outside their main specialism – for instance, to attain skills that they need or want.

Elda Bengu is a psychology student at Arizona State University in Tempe, an institution that has embraced AI tools through an AI Innovation Challenge in partnership with OpenAI. As part of that effort, Bengu’s lecturer Christiane Reves created a chatbot called Language Buddy that Bengu has used to practise speaking German. Research shows

“You can have back-and-forth conversations and ask clarifying questions, and that’s been very useful.”

that engaging in conversation is among the quickest ways to improve language skills, but students aren’t always able to say things in the clearest way or ramp up the complexity of their speaking. Language Buddy can.

Initially sceptical, Bengu says that after testing how the app handled her first language, Turkish, she was pleasantly surprised with its performance. “I was able to have a conversation and even ask it to speak in a certain dialect or accent, which would be especially useful if you’re learning to live somewhere new.”

Mastering the calendar

Another space in which students have successfully channelled the usefulness of generative AI is in streamlining their workflows. By using AI as a type of personal assistant, tools can quickly pull together information, prepare schedules and free up precious time.

Shivali Verma, who studies brain and mind science, and is splitting her time between University College London and Sorbonne University in Paris, uses chatbots as her go-to source for debugging data-analysis code. She doesn’t use them to write the code itself, but such tools have supplanted websites such as YouTube, Stack Overflow and GitHub, which she would previously have used to troubleshoot. “You can sit for hours looking through forums for a very specific question, or you can ask ChatGPT,” Verma says, noting how much time the practice has saved her. “Of course you need to be sceptical, but I would say most of the time the information is correct.”

Several students, including Ivich, have also used NotebookLM to create AI-powered podcasts on a topic – a sort of modern literature

review in which the chatbot synthesizes a handful of user-supplied papers to create a fictional conversation between two hosts. “It can get cringey, but it’s a fun way to engage with the material,” Ivich says, adding that it has helped her squeeze studying into other parts of her day. “If I’m about to take my dog on a walk, I’ll listen while I’m out.”

Jackson Frazier, who is studying aerospace engineering at Embry–Riddle Aeronautical University in Florida, has integrated ChatGPT into his Google calendars, and now asks the bot to make him a cohesive weekly schedule or to help with divvying up roles in a group project on the basis of each person’s strengths. “It has helped me become much more organized as a person, which is something I’ve always struggled with,” Frazier says.

Navigating life away from home

With chatbots integrating into so many facets of academic life, it hasn’t taken long for them to break into personal lives as well, says Leo Wu, who is studying economics and systems design at Minerva University and is president of a student-run group there called AI Consensus, which advocates for the ethical use of AI in education.

Although some experts warn that over-reliance on AI risks dampening social skills, students say that these tools, when they are used thoughtfully, have actually deepened their personal connections and helped them to lead healthier lives. Indeed, students have shared experiences of using chatbots to plan travel itineraries, create art, navigate painful relationship break-ups or difficult conversations with friends, or bridge the distance with home when they leave for the first time.

Frazier, for example, doesn’t just use ChatGPT to organize his class schedule but also to build in time for extracurricular activities and enforce some work–life balance. “I tell ChatGPT about myself, my hobbies and a list of goals I want to achieve,” such as exercising four days a week or learning a song on the piano, “and it will give me a layout of a plan to help me get there.”

And, because Verma attends university in France, she now schedules regular calls with friends back home in the United Kingdom. Sometimes they use ChatGPT to generate silly drawing prompts, one of which has become the design for a tattoo of the London skyline for Verma and her friend. Verma says that she once tried to cut the chatbot out of her life for both financial and environmental reasons, but found that it was simply too useful.

“I’ve come to depend on it,” Verma says. “I try to be mindful of how much I use them, and to make sure I’m using them responsibly, but I no longer see a future where I don’t use them.”

Amanda Heidt is a freelance journalist in southeast Utah.

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ILLUSTRATION: PABLO CARACOL

CHOOSING THE RIGHT AI TOOL FOR THE JOB

Curious about using artificial intelligence to boost your research? Here are the programs you shouldn't miss. **By Amanda Heidt**

When Mohammed Shafi, a PhD student in civil engineering at the Indian Institute of Technology Guwahati, first saw his friends testing out artificial intelligence (AI) tools back in late 2022, he didn't immediately see the appeal. Mostly, people seemed to be using generative AI platforms such as OpenAI's ChatGPT as a replacement for Google, or as a novelty for drumming up ideas

for practical jokes and pet names. "They were fun to play around with, but I didn't necessarily sense any relevance to my own coursework or my research," he says.

He quickly came around, however, when he started seeing more AI tools being built to meet the needs of students and scientists. Now a daily user of AI, Shafi has pieced together an entire pipeline of AI-powered platforms that feed into one another. These update him on

new research, break down complex topics, troubleshoot experiments, organize his writing and citations, and help him to navigate the demands of classes and research.

Shafi now says that the arrival of AI has been "a revolution for research", a sentiment seemingly shared by others. Surveys show that many university students and scientists are using AI in their work, often on a weekly or even daily basis. And whereas many educators and

Work/Careers

academic institutions initially responded with wariness, academia seems increasingly willing to allow students to use AI, albeit in controlled ways. Although it wouldn't be impossible to go back to the way he did things before, Shafi says, "it's hard to imagine wanting to".

Here, *Nature* explores how academics and students can harness AI to streamline various parts of the research process.

Sharpen your literature review

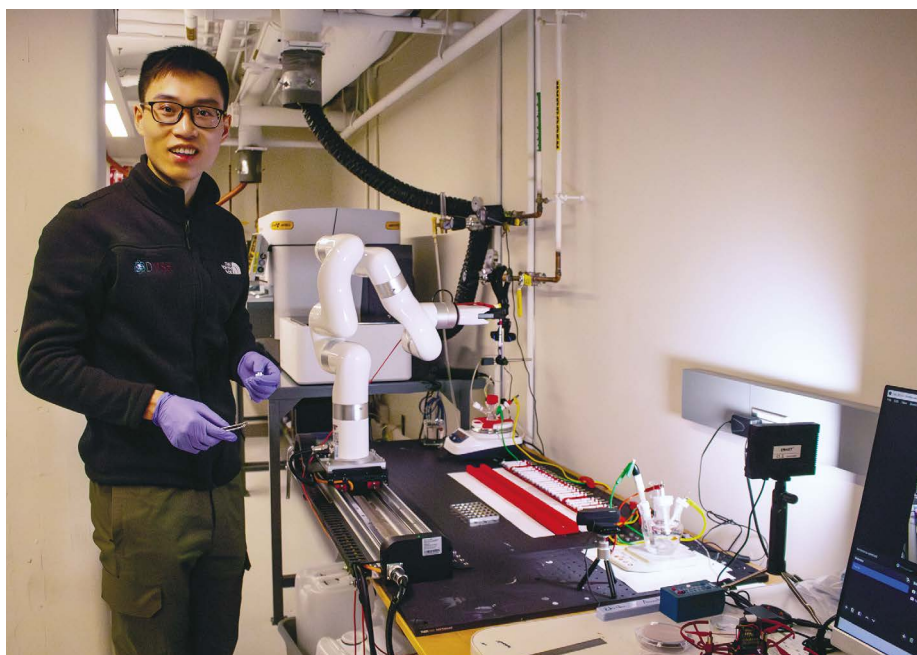
Daniel Weld, chief scientist at the academic search engine Semantic Scholar, who is based in Seattle, Washington, says that many popular AI platforms have "advanced enormously" in an area called active learning – a method that mimics how a person would approach a research question. Programs such as Google's Gemini Deep Research and OpenAI's Deep Research offer the most powerful tools in this regard, and many companies are launching similar products.

Students can enter a query, supported by their own data or documents, and then step away as these advanced models conduct in-depth searches over 30 minutes or so. The final report might include text, figures and visualizations, and all output is thoroughly referenced – another jump over past iterations, says Isa Fulford, a technical staff researcher at OpenAI in San Francisco, California, who helped to develop Deep Research. "Especially in the context of scientific research, we recognize that veracity is critical, and we think this model is better at including the proper citations than any other model we've released," she says.

Chuck Downing, a PhD student in accounting at the Massachusetts Institute of Technology (MIT) in Cambridge, says that these deep-research tools have been especially useful when digging into unfamiliar topics. During one project, Downing used OpenAI's Deep Research to create a report ranking various approaches for reducing emissions at manufacturing plants. "I didn't know much going in, but I learnt quite a bit, and so I use these deep dives all the time now," he says. "It's better than anything else I've used so far at finding good papers and in presenting the information in a way that I can easily understand."

Other programs enable students to delve more deeply into a single document or small collection of papers. The student-focused AI platform SciSpace, for example, has a 'Chat with PDF' function. Users can upload a paper and ask questions about its content – a feature shared by other platforms, such as Claude, NotebookLM and PDF.ai.

For David Tompkins, a PhD student in human development at Cornell University in Ithaca, New York, this approach has helped him to stay on top of the burgeoning scientific literature. Tompkins often goes to journal-club meetings having used Claude to generate a summary of a chosen paper, which he then follows up with



Researcher Zhichu Ren created an AI-based system called CREST to run experiments.

more-targeted questions based on the group's discussion. "I'm still a big believer in actually reading papers to fully understand them, but it's become much easier to do my prep when I'm feeling stretched," he says. "In some ways, I feel I'm engaging more with the material through these tools than I did before them."

Create your hypothesis

The ability of AI to pull together many threads of information has seemingly made it easier to identify research gaps and connect ideas – although a recent survey suggests that an over-reliance on generative AI could dampen a person's critical-thinking skills (see go.nature.com/4tjv7vq). Weld, who is also an AI researcher at the Allen Institute for AI in Seattle, says there has been so much demand for tools that assist with ideation that he and his team are developing hypothesis-generation and detection products that attempt to combine ideas across papers into something new. "We have them running internally, but we're trying to make sure that they're working robustly," he says, adding that his Allen Institute group hopes to release them publicly in the next few months.

Shafi turned to programs such as the visualization tool Research Rabbit when working on his dissertation, which focuses on how microplastics are transported through soil and into groundwater. Research Rabbit takes a single 'seed paper' and generates an interconnected web of research linked by topic, author, methodology or other key features. By piping its results into a chatbot such as ChatGPT, "it's possible to query the body of work for hidden links or new ideas", Shafi says.

AI-powered programs are also proving increasingly capable as experimental assistants. As a PhD student at MIT, Zhichu Ren created the

software Copilot for Real-world Experimental Scientist (CRESt), which combines several AI technologies into an enhanced chatbot (Z. Ren *et al.* Preprint at ChemRxiv <https://doi.org/pdwv>; 2023). Users can chat with CRESt as they would with a colleague, and it can help to craft and run experiments by retrieving and analysing data, turning equipment on and off using digital switches, powering robotic arms, documenting findings and alerting scientists by e-mail when issues arise or protocols end (see go.nature.com/3fgdkrg). In a 2023 conference paper, CRESt assisted researchers by prioritizing candidate alloys for a new fuel cell, and suggested experiments that the group might run to test them (see go.nature.com/4cdn2tr). "I wanted to create a tool that can continue to help even as your needs change," says Ren, who now works at the AI start-up firm Labig in Cambridge, Massachusetts. "AI can do that in a way that static, written documentation cannot."

But even for students who do not have access to something as advanced as CRESt, AI can still function as a helpful colleague. Gemini Deep Research, for example, can generate a "personalized multi-point research plan" among other features, and resources such as Scite and Elicit are billed as research assistants. Users can give these programs a handful of papers or a working hypothesis, for example, and ask for a set of experiments to test the theory.

Joseph Fernandez, a PhD student in biomedical engineering at the University of Colorado Anschutz Medical Campus in Aurora, says he continues to use ChatGPT for most things, including troubleshooting his experiments. In the past, the bot has helped him to brainstorm explanations when one of his assays was returning unusual values, and to calculate serial dilutions to avoid wasting expensive reagents.

JASON SPARAPAN/MIT DEPT OF MATERIALS SCIENCE AND ENGINEERING

ChatGPT has also served as a stand-in for his committee, generating pointed questions to test his research proposal before his exams.

"I think you're really only limited by your imagination, even if some uses are more mundane than others," he says. "Nowadays, if a question or task pops into my mind, I'm generally wondering if ChatGPT can help with it."

Streamline your statistics

Code editors including GitHub's Copilot, Amazon Code Whisperer (now Amazon Q Developer) and AnySphere's Cursor aim to make it easy for beginners to use coding to organize data, create analysis pipelines, run descriptive statistics and generate visualizations. Such tools, researchers note, have also largely overtaken websites such as GitHub and Stack Exchange as the main resources for troubleshooting. Rather than spending hours looking for answers, users can simply highlight a section of code and ask a chatbot to fix it, Downing says.

"Thinking about what a PhD student primarily does, the largest tasks are increasingly coding and data analysis, at least for computational fields, so anything that helps there is just disproportionately useful," he says. Although he already considered himself an adequate coder, he says that his preferred tool, Cursor, has made him better by removing the more tedious aspects and making it easier to probe specific aspects of a data set. Instead of spending all of his time debugging (cleaning the code), he says, he is "putting more effort into really getting to know the underlying data and engaging with my code in ways that help me learn. If I get curious about something, it's very easy to generate descriptive statistics, something like a chart."

Tompkins has likewise found tools such as Claude to be essential for writing code for compelling, dynamic visualizations. Creating a good graph, particularly if it's interactive, can require hundreds of lines of code, and Tompkins says that, in the past, that level of effort had put him off. "But once I started using Claude, I was able to have it write out those literal hundreds of lines of code," he says. The resulting visualizations have gone a long way towards helping others to understand his research, which describes how small changes in the way in which people experience information can drive their reactions to it.

He adds, however, that he still always writes his own code for statistical analyses: "I want to make sure that whatever I'm reporting is something that I fully understand and can stand behind when I submit a paper."

These AI programs focus on generating new code, but Gaurav Ragtah, who founded a platform called CatalyzeX, saw an opportunity to repurpose existing code. If researchers write a new analysis pipeline for each experiment, for example, it can make it more challenging for others to reproduce their work, particularly if the documentation is poor or a developer

stops updating their code. Instead, Ragtah, who is based in San Francisco, wanted to make it easier to locate and share code that others have published. CatalyzeX uses a web platform and browser extension to flag open-source code shared in papers indexed on websites such as Google Scholar or PubMed, and researchers can search for code through the platform using keywords. Someone interested in using machine learning to aid in cancer

"As amazing as some of these code generators are, we don't need to reinvent the wheel every time."

detection, for example, could search for a modified data-processing pipeline that helps to address the fact that publicly available data often involve small sample sizes.

"As amazing as some of these code generators are, we don't need to reinvent the wheel every time when there are excellent examples of what you're trying to do," Ragtah says. "Open source keeps people from having to start from scratch and gives them a scaffold on which to build and improve, while making research more easily comparable."

Polish your writing

The earliest, most obvious uses for chatbots were as writing tools, and text generation remains a top task. Generalized AI can sometimes struggle with the intricacies of scientific writing, however, so developers are creating platforms that appeal directly to the unique needs of researchers and students in scientific, technical and medical disciplines.



María Mercedes Hincapié-Otero.

Tools such as Paperpal and Thesify check academic manuscripts against journal submission guidelines, for example, and offer templates for research proposals, literature reviews, abstracts, dissertations and essays. SciSpace, along with platforms including Coral AI, Quillbot and OpenAI's Whisper, also advertise translational capabilities covering dozens of languages — a useful tool for researchers such as María Mercedes Hincapié-Otero, a PhD student in cell biology at the University of Helsinki in Finland, who grew up speaking Spanish in Colombia.

Hincapié-Otero often uses AI to check her writing for flow, grammar and tone, and has used ChatGPT, among other programs, to draft informal e-mails and to apply successfully for jobs. "There's already a disproportionate burden on scientists who speak English as a second language," she says, adding that she has felt self-conscious at times over her ability to explain complex scientific topics when writing in English. "AI makes a great stand-in for when I want to run my ideas or my writing by another person, but no one is available, and it just generally gives me peace of mind to focus on more important things."

This idea of increasing access to institutional knowledge is part of what motivated Marc-Oliver Gewaltig, a former neuroscientist, to co-found the Swiss company Thesify in 2024 as a multi-stop platform for students. "I had the advantage of an extremely attentive supervisor who read everything carefully and gave us feedback," he says. "That's not always the case, and I have the added perspective now of seeing how hard it is for faculty members to even find the time to pass on all their knowledge to students. Fortunately, we can ease that burden on both sides with AI."

As well as providing feedback on research papers and dissertations, Thesify is one of the only platforms to focus on training students in another essential space: grant writing. Grant proposals, Gewaltig says, are rarely represented in AI training data, yet learning how to structure them is a necessary skill that typically takes years to cultivate. Similarly, the platform's conference and journal finders help students to identify meetings and publications where their work is most likely to be accepted.

"The goal of this tool, which is really the goal of many of these platforms, is to increase efficiency and productivity, but in science that sometimes means something different," he says, referencing the fact that so much of being successful in academia comes down to experience and good mentorship. "There's still a long way to go in creating something like truly generalized intelligence, but we have the capabilities now to make information more available to everyone."

Amanda Heidt is a freelance writer and editor in Moab, Utah.



ILLUSTRATION: ANTONIO RODRÍGUEZ

AI AND THE PHD STUDENT: FRIEND OR FOE?

Graduate students increasingly use AI to draft, code and search – but many fear it could erode the very skills they are trying to build. **By Linda Nordling**

Leona Diala uses artificial-intelligence tools for many aspects of her PhD: to search the literature, design presentations, generate code and summarize papers. The tools save her time and give her ideas, she says. “I would say AI is a blessing for researchers today.”

At the same time, Diala, who is studying infectious-disease modelling at the University of Abuja, is concerned that AI overuse is undermining essential academic skills. She checks every reference and fact that AI tools give her and rewrites their text in her own words. But she worries about the next generation of researchers. When she was an undergraduate, she says, “there was no AI like this. We sat for hours and read, practised, tried and retried

until we got it. Now people want AI to write everything.” She adds: “AI is a blessing, but it has made students lazy. People don’t go the extra mile to build the skill.”

Diala’s ambivalence is typical of PhD students. When *Nature* surveyed almost 3,800 PhD students last year, three-quarters thought AI tools could help students to work more efficiently, and 71% felt it was acceptable to use them to support their studies – yet the majority also voiced strong concerns. Some 81% said they don’t fully trust AI tools and 65% worried that AI weakens thinking, research and writing skills.

Since ChatGPT launched in November 2022, AI use has exploded across higher education. In a survey of 1,041 UK undergraduates published

in February 2025, 88% admitted to using AI for assessments, up from 53% the year before (see go.nature.com/4d37rcc). The proportion of respondents who had used any AI tool also jumped, from 66% in 2024 to 92% in 2025. Such a rapid change in behaviour is “almost unheard of”, wrote study author Josh Freeman, policy manager at the Higher Education Policy Institute in Oxford, UK, in a statement accompanying the results.

Doctoral students are now charting paths through territory their supervisors never had to navigate. Some use AI daily and swear by it; others refuse to touch it, worried about the cost to their development as researchers. Most fall in between, working out their own rules for when AI helps and when it hinders.

“AI can be your greatest ally or your worst enemy. It all depends on how you use it,” says Yinghui He, a PhD student at Tsinghua University in Beijing who uses computer methods to study microorganisms in the deep ocean. She turns to AI tools such as ChatGPT and Gemini every day, as well as Chinese AI tools, for tasks including checking grammar in her written English and generating code. “The AI writes code faster than me, and saves me a lot of time,” she says. “But once it’s generated the code, I need to check if it’s right. That is very important.”

That’s a lesson that Richard Ang, a PhD student in soil microbiology at the University of Western Australia in Perth, learnt the hard way. He once asked ChatGPT to calculate fertilizer doses for an experiment. When the experiment failed, he asked the tool to show its thinking. “It totally misunderstood my question,” he says. “AI will never tell us that our design is uncommon, or wrong,” he adds. “If we ask AI to carry out a ridiculous or impossible task, it will do it.” Now, he always asks the tool to explain its reasoning step by step, and he cross-checks answers using multiple tools.

Diala learnt the same lesson when she plotted a graph and asked AI to explain it. “It said a value increased when the graph showed a decrease,” she says. “Before we use AI, we should have some knowledge of what we’re asking, so we can catch mistakes.”

Where to draw the line

The confusion many PhD students feel comes partly from AI’s technological acceleration outpacing the ability of universities to govern it. According to a report published in January by the European University Association in Brussels, which surveyed universities across Europe about their policies for doctoral education, only 5% felt the existing AI guidelines were sufficient (see go.nature.com/4stfdp). The largest group, 38%, reported that they were still establishing policies for the first time, and 13% had no policies at all.

In the absence of clear direction, students are turning to educators such as Amina Yonis for answers about how to use these tools responsibly. Yonis (who trained as a molecular biologist at University College London) now runs The Page Doctor, an academic-support company that helps students with writing and editing. She has built a YouTube following of almost 500,000 subscribers by offering practical tips for navigating academic life, including how to use AI ethically in a way that doesn’t harm intellectual development.

“There are parts where I’d say ‘use AI’ and parts where I’d say ‘don’t,’” says Yonis, who is based in Dubai, United Arab Emirates. Tasks such as finding, digesting and organizing research literature fall in the ‘use AI’ column. “It helps you read 50 papers in a month rather than 50 in a year.” But she urges students to



Amina Yonis offers tips on ethical AI use.

avoid the tools for data analysis, to stay in control of the intellectual work and because of data protection: many AI tools do not keep what you tell them confidential.

Yonis also urges caution in using AI to write a thesis. The technology is useful for coming up with a structure or cleaning up text, she says. But getting it to compose long sections from scratch can be problematic. “If you let AI write and then you look, you’ll struggle to break away from the example,” she says. “It’s better to write it yourself and then use AI to clean up.”

For language help, however, AI can be transformative. Manikandan Palanichamy, an electrical and computer engineer at Østfold University College in Fredrikstad, Norway, who supervises PhD students, says that it gives those who don’t have English as their first language access to cutting-edge research and helps them to express complex ideas more effectively, which “democratizes PhD training significantly”. However, although AI can be used to amplify thinking, it should not replace it, he says. “As a professor, my challenge for PhD or master’s or bachelor’s supervision is teaching students this distinction early.”

Not all AI tools are created equal, Yonis says. She recommends stepping away from generic AI tools and exploring applications that are purpose-built for academics, such as Paperpal for writing or Consensus for literature searches (see ‘AI joins the lab’).

What skills will matter?

Even with practical guidance on how to use AI responsibly, deeper questions remain. As the technology hurtles forwards, it’s difficult to say precisely what skills PhD students need to hone or protect to navigate the future.

Natalia Bielczyk, a computational neuroscientist who splits her time between California, Poland and the Netherlands and founded Ontology of Value, a company that helps professionals to navigate career transitions, says

AI joins the lab

A growing range of artificial-intelligence tools caters specifically to academics.

More AI apps are being purpose-built for research. Most have tiered pricing models, with free limited-use options or free trial periods. We highlight a selection.

For discovering and understanding literature, ResearchRabbit maps citation networks and helps to find connected papers. The tool Elicit searches a large body of academic papers and can extract key findings, summarize papers and answer research questions. Consensus returns evidence-based answers with linked citations. Scite looks at how other papers talk about a study — whether they support, contradict or simply mention its findings. Connected Papers, Litmaps and Inciteful (each has a free version) create visual maps of research fields and paper relationships.

For data analysis and coding, Julius analyses data sets through language prompts, and generates code and visualizations. Bohrium is a cloud platform for running scientific simulations, notebooks and managing research data, and GitHub Copilot suggests ways to complete your code as you type.

For writing and editing, Paperpal offers writing suggestions and language polishing specifically for research papers. The Prism workspace from ChatGPT creator OpenAI, based on the model GPT5.2, lets users refine documents written in LaTeX, a document-preparation system used for some scientific papers.

Remember, whichever tools you use, they should help you to learn — not replace your thinking. Never upload sensitive, proprietary or identifiable participant data to AI tools without proper data-use agreements and institutional approval, and don’t rely on AI-generated code without understanding and testing it thoroughly. Don’t use AI to write entire papers or manuscript sections without disclosing what you’ve done — this might violate academic-integrity policies.

Finally, if you feel you are becoming too dependent on AI systems, Amina Yonis at academic-support company The Page Doctor has some simple advice: go back to conventional methods. “AI has only been around for about three years. Most people finished PhDs without it. You can do it.”

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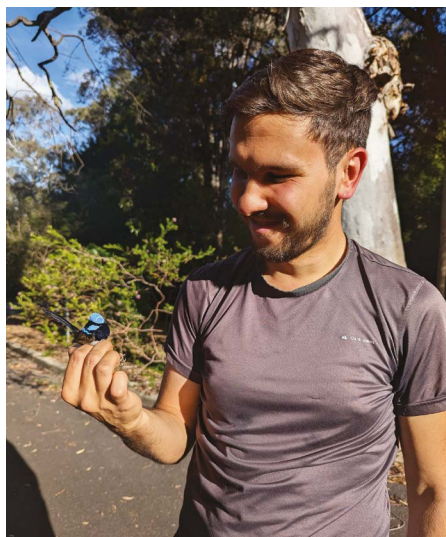
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Samuele Ramellini with a superb fairywren.

PhD students might want to think about their value differently in the age of AI. Rather than competing with AI on speed or recall, doctoral students should lean into areas that machines still struggle with: framing good questions, navigating ambiguity and designing ways to test ideas in the real world.

“The superpower of the PhD is that it gives you a systematic approach to solving problems,” she says. AI can help with routine tasks such as scanning the literature or polishing drafts, and so leave more time for the kind of slow, conceptual work that underpins real breakthroughs. “It’s unlikely that AI will replace higher-range thinking any time soon,” Bielczyk suggests. “If anything, it raises the bar for humans to specialize in the parts of research that only humans can do.”

But to establish good habits, students must think about these issues long before they reach doctoral level, says Palanichamy. “Students need proper ethical guidance on responsible AI use from their school years onward,” he says. “Otherwise, we risk producing a generation of researchers who can operate AI tools but lack the fundamental skills to push boundaries and create genuine breakthroughs.”

PhD students must also learn skills, such as the ability to ask precise questions and check AI-generated answers, says Alex Sen Gupta, a climate scientist based at the University of New South Wales in Sydney, Australia. Last November, he published an essay calling for an overhaul of the PhD, from its duration to how it’s assessed and taught (see *Nature* 647, 27–28; 2025).

“If used well, AI can certainly massively enhance PhD capabilities. It frees up much more time to reflect on ideas, questions, solutions, translation to real-world impacts, big-picture stuff,” says Sen Gupta. But concerns about AI eroding skills are legitimate, he adds. Students who use it as a shortcut to avoid thinking will emerge with serious gaps

in expertise, whereas those who use it to push their thinking deeper could thrive.

The problem, Sen Gupta says, is that nobody can predict how the field will develop. “It could all change a year from now, or two years from now, or five years from now.” This uncertainty makes it difficult to know which skills will remain essential. Meanwhile, technology companies are attempting to inject AI across every cognitive domain, he says. “Every time we find a gap that it can’t do, it’s going to train on that gap.”

Opting out entirely

Not everyone is convinced that maintaining this balance is worth the effort.

In November, Samuele Ramellini used AI to help him plot a graph for the first time. “I’ve always coded myself. I learnt it the hard way – courses, long hours,” says Ramellini, who is halfway through his PhD researching the complex social behaviour of a bird called the superb fairywren (*Malurus cyaneus*) at the Australian National University in Canberra.

The AI worked well, he says. But a couple of weeks later, he couldn’t remember the code it had produced, or even what he had asked the AI to do. By contrast, he remembers exactly what he did for each plot in his undergraduate thesis almost ten years ago. “That’s what frustrates me about AI. It gives immediate reward, but you don’t learn or improve.”

“We risk producing a generation who can operate AI tools but lack the fundamental skills.”

As for putting his research ideas into AI tools, Ramellini gives an emphatic no. “Absolutely not. ChatGPT isn’t getting my ideas. I’m old-school: paper, Post-its on the floor, whiteboards. It’s much better.” He would rather brainstorm with his supervisor, and says that AI removes some of the fun of research. “I find my job extremely entertaining. I studied for years to get here. Now I would leave it to AI to save an hour and then what? Go on Facebook and look at AI pictures?”

There are probably others like Ramellini, and Bielczyk thinks that the current AI frenzy could lead some people to resist the technology entirely. But she also expects that competition for grants and jobs will tempt many to use AI tools in dubious ways, taking shortcuts to advance their careers. Those in between will get better at filtering information and understanding when the technology can and should be used, she says. “I think in five to ten years, it will be more obvious which tasks we can leave to AI, and which we have to do ourselves.”

Linda Nordling is a freelance journalist in Cape Town, South Africa.

FRANZISKA HACKER